Bug Tracking Project

Problem Statement #25: Municipal Solid Waste Route Optimizer

Name: Merull Shah

SRN: PES1UG24CS270

Problem Statement: BPS#25

Jira Link: https://merullshah25.atlassian.net/jira/software/projects/BUG/

1. Bug Tracker Project Creation

A Jira Bug Tracking project was created to log defects found while testing the Municipal Solid Waste Route Optimizer against its Lab 1 acceptance criteria. Each defect references the requirement whose acceptance criterion it violates.

Field	Value
Project name	BugTracker_BPS#25
Project key	BTBPS25
Template	Bug tracking (Jira Software)
Work item type	Bug
Workflow	Open → In Progress → Resolved → Closed
Defects logged	9

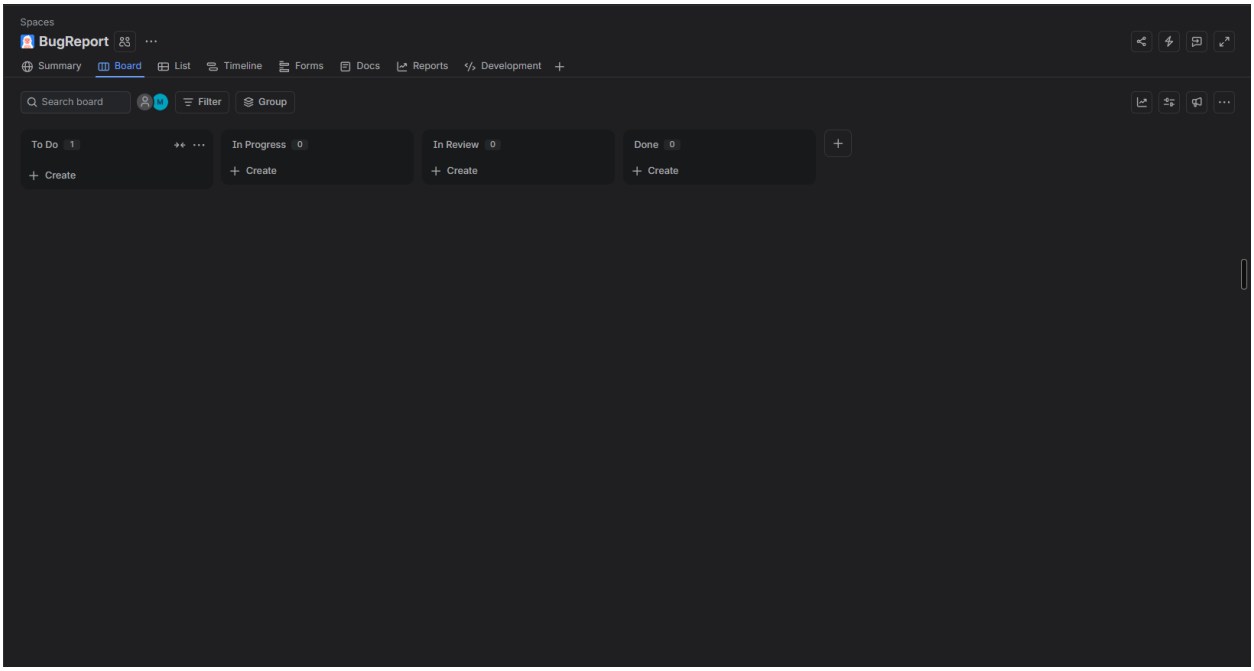


Figure 1: Bug Tracking project created in Jira

2. Defect Summary

The table below lists every defect raised, its severity and priority, and the requirement whose acceptance criterion it breaches. Severity describes the impact on the system; priority describes the urgency of the fix.

Bug ID	Summary	Severity	Priority	Violates
BUG-01	Bins at exactly 80% fill are excluded from the generated route	Major	High	FR-001
BUG-02	Manually tagged bin is dropped when the route is regenerated	Major	High	FR-002
BUG-03	Collected bin still appears on the driver manifest after confirmation	Major	High	FR-003
BUG-04	Complaint accepted without a valid bin reference when submitted offline	Major	Medium	FR-004
BUG-05	Breakdown alert not raised when the truck loses GPS before reporting	Critical	Highest	FR-005
BUG-06	Route recalculation exceeds the 45-second target at full scale	Critical	Highest	NFR-001
BUG-07	Telematics payload written to the application log in plaintext	Critical	Highest	NFR-002
BUG-08	Stale bin readings older than 24 hours are not rejected	Minor	Medium	FR-001
BUG-09	'No collection required' message not shown for an empty zone	Minor	Low	FR-002

- Critical (3): BUG-05, BUG-06, BUG-07 — service continuity, performance and security failures.
- Major (4): BUG-01 to BUG-04 — core collection workflow defects with visible user impact.
- Minor (2): BUG-08, BUG-09 — data-freshness and messaging defects with workarounds.

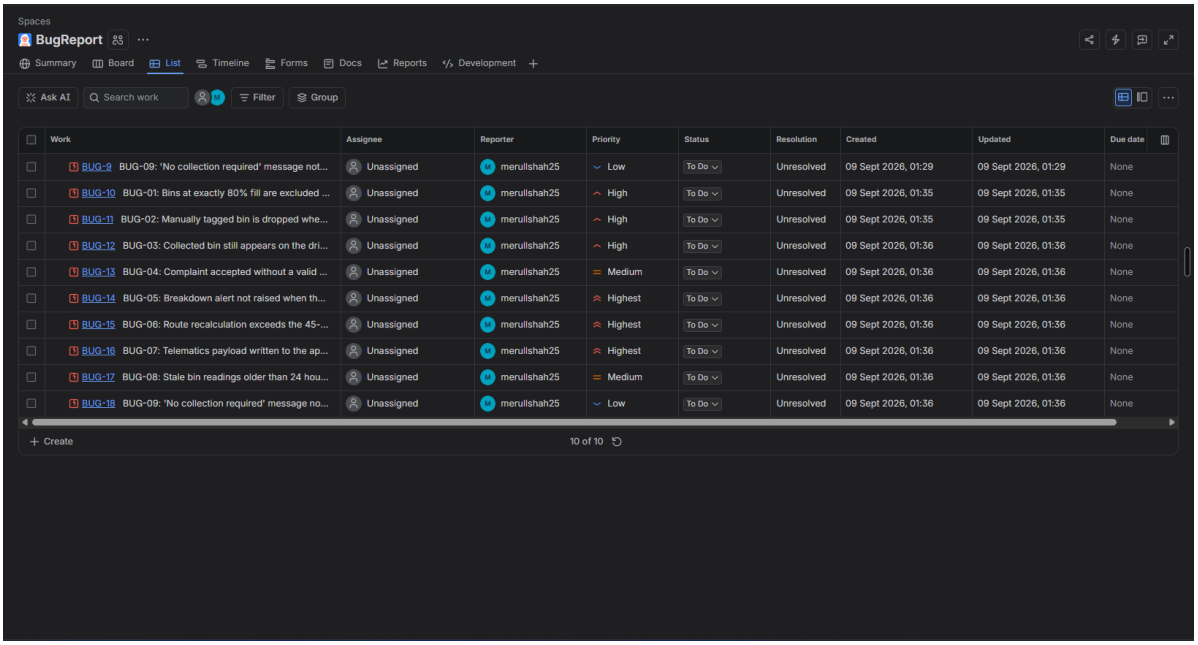


Figure 2: All defects reported, shown in the Bug Tracker list view

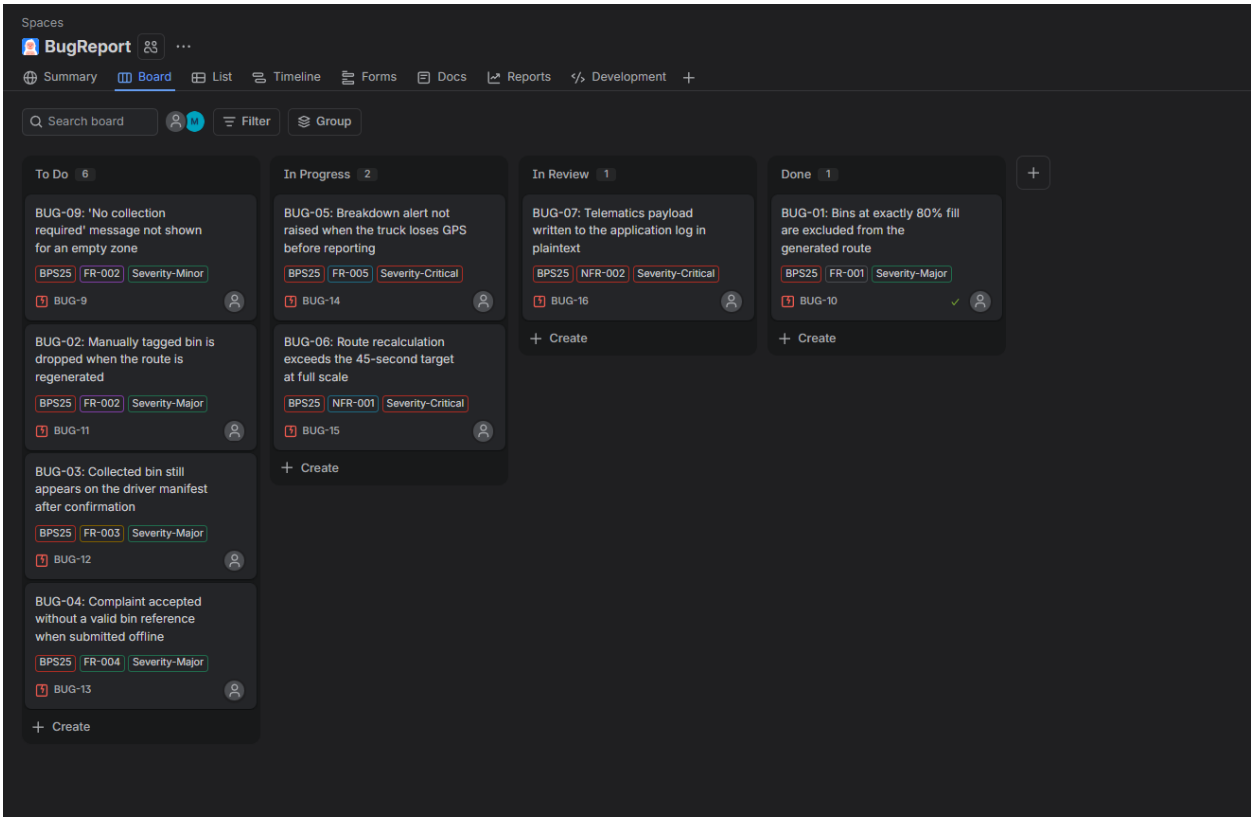


Figure 3: Bug board with defects distributed across workflow columns

3. Defect Details

Each defect below is recorded in Jira with the same fields: steps to reproduce, expected result and actual result, plus the requirement it traces to.

BUG-01 — Bins at exactly 80% fill are excluded from the generated route

Field	Detail
Severity / Priority	Major / High
Violates requirement	FR-001
Steps to reproduce	Set a bin's fill level to exactly 80%. Generate a route for that zone.
Expected result	Bin is included — the requirement states at or above the 80% threshold.
Actual result	Bin is excluded; the filter uses a strict greater-than comparison.
Status	Open

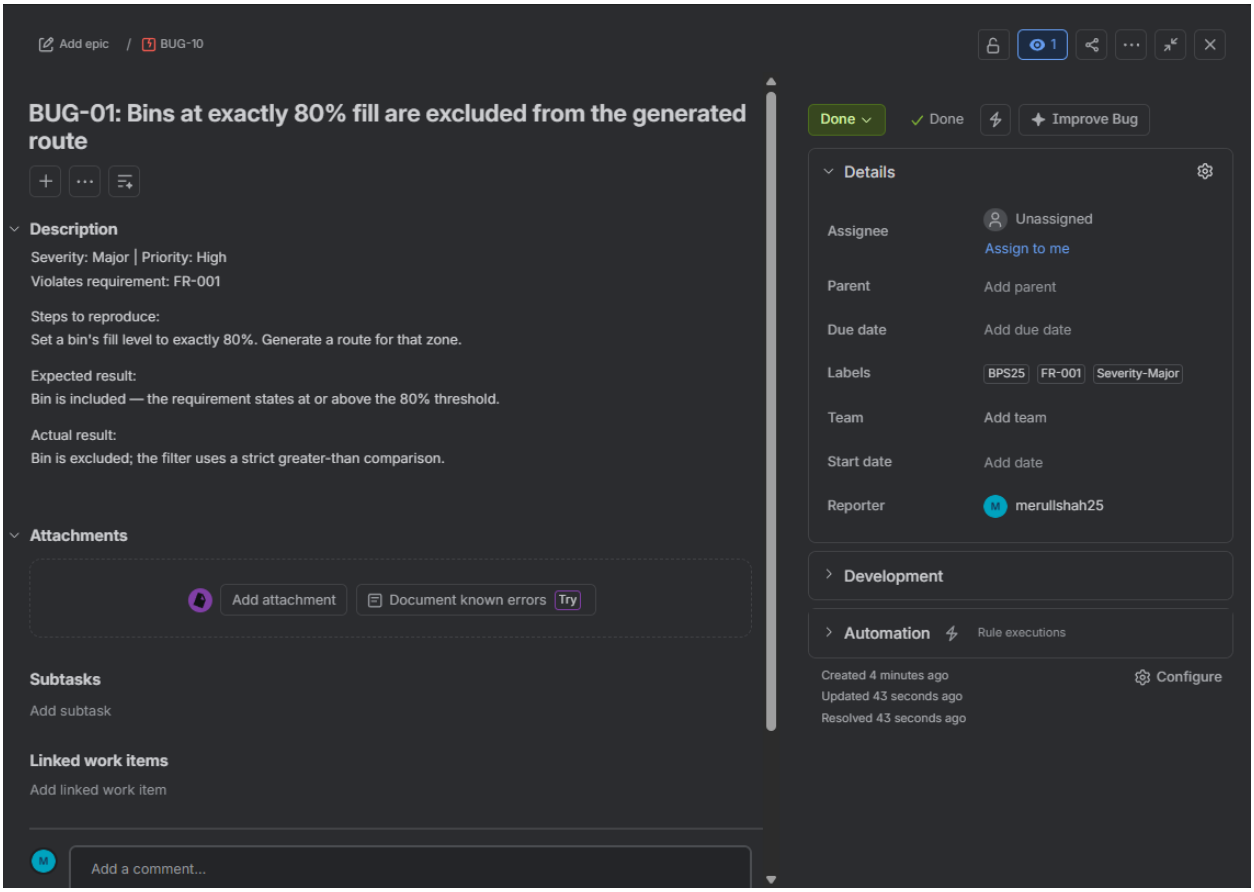


Figure 4: Jira detail view for BUG-01

BUG-02 — Manually tagged bin is dropped when the route is regenerated

Field	Detail
Severity / Priority	Major / High
Violates requirement	FR-002
Steps to reproduce	Tag a bin below 80% for pickup. Generate a route, then regenerate it without publishing.
Expected result	The tagged bin remains on the manifest across regenerations.
Actual result	The tag is cleared on regeneration and the bin disappears from the manifest.
Status	Open

BUG-03 — Collected bin still appears on the driver manifest after confirmation

Field	Detail
Severity / Priority	Major / High
Violates requirement	FR-003
Steps to reproduce	As a driver, mark an assigned bin collected. Observe the manifest for 10 seconds.
Expected result	Bin status resets to 0% and it leaves the route list within 5 seconds.

Field	Detail
Actual result	Status resets in the backend but the manifest only refreshes on manual pull-to-refresh.
Status	Open

BUG-04 — Complaint accepted without a valid bin reference when submitted offline

Field	Detail
Severity / Priority	Major / Medium
Violates requirement	FR-004
Steps to reproduce	Put the citizen app in airplane mode. Submit a complaint with the bin field blank. Reconnect.
Expected result	Submission is rejected; a bin reference and timestamp are mandatory.
Actual result	The queued complaint syncs and is logged with a null bin reference.
Status	Open

BUG-05 — Breakdown alert not raised when the truck loses GPS before reporting

Field	Detail
Severity / Priority	Critical / Highest
Violates requirement	FR-005
Steps to reproduce	Simulate a telematics breakdown event from a truck with no current GPS fix.
Expected result	Supervisor is alerted within 30 seconds and the route is recalculated.
Actual result	The event is discarded during validation; no alert fires and the bins stay unassigned.
Status	Open

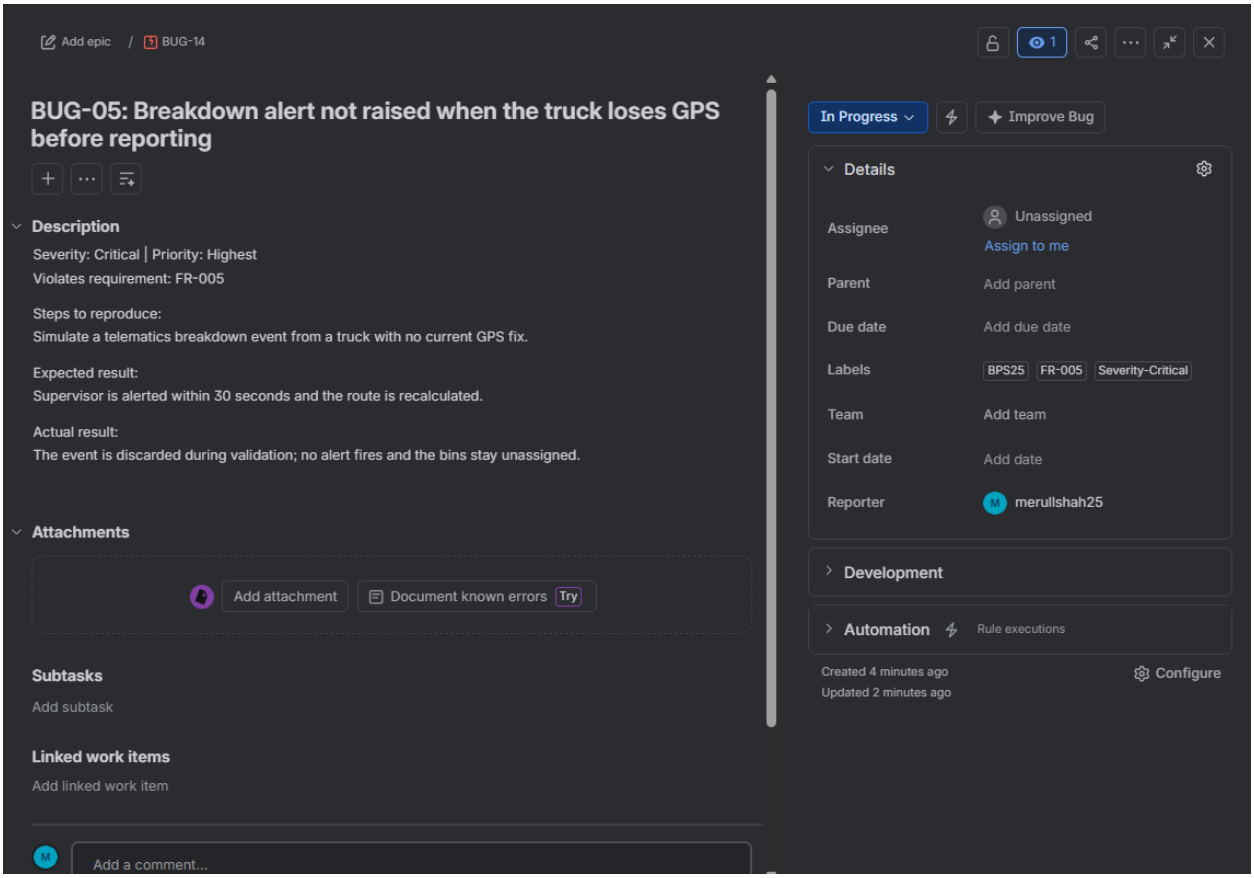


Figure 5: Jira detail view for BUG-05

BUG-06 — Route recalculation exceeds the 45-second target at full scale

Field	Detail
Severity / Priority	Critical / Highest
Violates requirement	NFR-001
Steps to reproduce	Run the benchmark harness with 50 trucks and 10,000 bins under simulated peak load.
Expected result	Recalculation completes in under 45 seconds.
Actual result	Recalculation takes appreciably longer than the target; record the measured figure here.
Status	Open

BUG-07 — Telematics payload written to the application log in plaintext

Field	Detail
Severity / Priority	Critical / Highest
Violates requirement	NFR-002
Steps to reproduce	Enable debug logging and trigger a telematics position update. Inspect the application log.
Expected result	No location or telematics data appears unencrypted in any store.

Field	Detail
Actual result	Raw coordinates are written to the log file, bypassing the encryption-at-rest control.
Status	Open

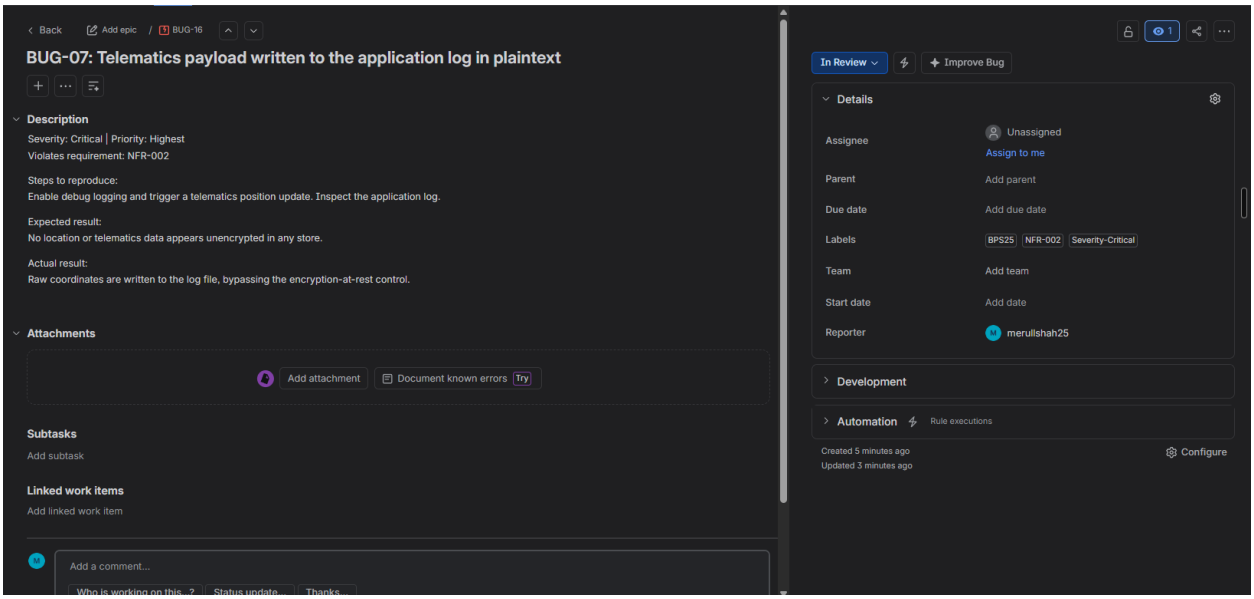


Figure 6: Jira detail view for BUG-07

BUG-08 — Stale bin readings older than 24 hours are not rejected

Field	Detail
Severity / Priority	Minor / Medium
Violates requirement	FR-001
Steps to reproduce	Set a bin's last reading timestamp to 30 hours ago with a 95% fill value. Generate a route.
Expected result	The stale reading is rejected as it breaches the use-case precondition.
Actual result	The bin is routed on the stale value, sending a truck to an already-empty bin.
Status	Open

BUG-09 — 'No collection required' message not shown for an empty zone

Field	Detail
Severity / Priority	Minor / Low
Violates requirement	FR-002
Steps to reproduce	Select a zone where no bin is above 80% and none is tagged. Click Generate Route.
Expected result	A 'No collection required' message is displayed (alternate flow 3a).
Actual result	An empty route manifest is generated and published instead.

Field	Detail
Status	Open

4. Defect Distribution

Grouping the defects by the requirement they violate shows where testing found the most fragility. The two nonfunctional requirements each produced a Critical defect, which is characteristic of quality attributes that are easy to state and hard to verify without dedicated tooling.

Req ID	Defects	Highest severity	Defect IDs
FR-001	2	Major	BUG-01, BUG-08
FR-002	2	Major	BUG-02, BUG-09
FR-003	1	Major	BUG-03
FR-004	1	Major	BUG-04
FR-005	1	Critical	BUG-05
NFR-001	1	Critical	BUG-06
NFR-002	1	Critical	BUG-07

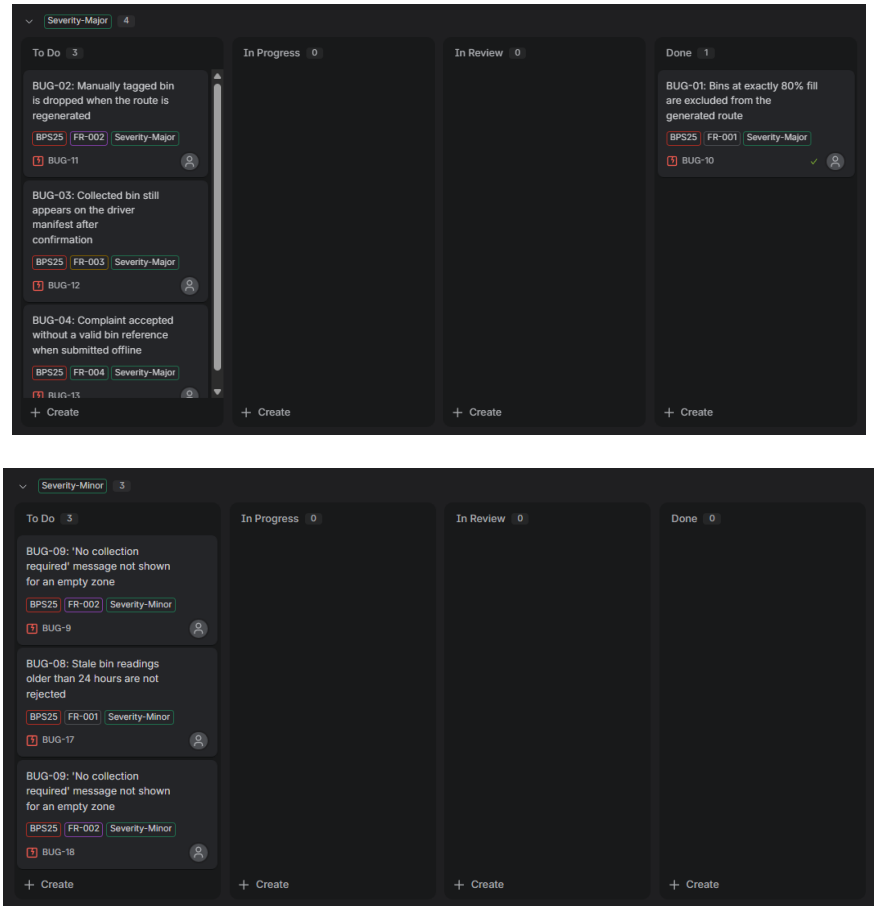


Figure 7: Defects grouped by priority or severity